Yifan Zhu

Toronto, ON | 647-554-9525 | zyfeleven@gmail.com | github.com/zyfeleven | linkedin.com/in/yifan-zhu-a8958a21b

PROFILE

Recent MSc Computing graduate with full-stack and cloud engineering experience across React, Node.js/Python services, PostgreSQL, and GCP serverless back ends. Builds LLM-enabled document-processing and agentic optimization workflows with a deterministic-first mindset: AI proposes, code verifies.

EDUCATION

Queen's University - MSc Computing, GPA 3.92/4.0

Kingston, ON | Sep 2024 - Apr 2026 | Thesis: University Teacher Assignment via MILP, heuristics, and LLM approaches

Brock University - BSc (Honours) Computer Science, First Class Honours

St. Catharines, ON | Sep 2021 - Apr 2024 | Dean's Honour List; Brock Returning Scholars Award 2022 and 2023

EXPERIENCE

Wizewerks Inc. - Full-Stack / AI Developer Intern

Toronto, ON | Mar 2025 - Aug 2025

- Built a production document-processing pipeline across React, REST APIs, Cloud Run, Cloud Functions, and Pub/Sub, orchestrating ingestion, classification, structured extraction, retry logic, and monitoring.
- Combined Google Document AI OCR with LLM-based classification and extraction to automate manual document review and turn ambiguous business requirements into a working cloud workflow.
- Refactored data models with Prisma and PostgreSQL, adding validation rules and schema changes that supported evolving business and audit requirements.

SELECTED PROJECTS

CIFR - Agentic Multi-Model LLM Framework for Constraint Satisfaction

Python, OR-Tools CP-SAT, pandas, Claude/OpenAI/Gemini APIs, React

- Designed a checker-driven iterative repair framework where LLM agents propose teacher assignments and deterministic checkers return feasibility violations for repair.
- Benchmarked MILP, genetic algorithms, simulated annealing, tabu search, and multi-LLM agents on two years of real CS-department data across feasibility, preference satisfaction, consistency, and fairness.
- Built a hybrid LLM-to-MILP pipeline using LLM outputs as deviation anchors, reducing MILP repair time from about 2 hours to about 1 minute while preserving feasibility guarantees.

Mise - Kitchen Inventory Mobile App

React Native (Expo), SQLite, Node.js, GCP Cloud Run, Qwen2.5-VL | 2026 - present

- Building a local-first bilingual kitchen inventory app with a stateless Cloud Run AI proxy for multimodal photo inventory intake, consumption logging, recipe structuring, and shopping-list categorization.

Graph Clustering for Vehicular Edge Networks

PyTorch, PyTorch Geometric, scikit-learn, Python | github.com/zyfeleven/veins_with_GNN

- Built a Graph Autoencoder clustering pipeline with SAGEConv, trained via link prediction and evaluated against scikit-learn baselines using silhouette, Davies-Bouldin, and adjusted Rand index.

PUBLICATION

Second author, *Vision Language Models for Optimization-Driven Intent Processing in Autonomous Networks*, accepted at IEEE ICC 2026; arXiv:2601.12744. Benchmarks VLMs for generating optimization code from network diagrams across IntentOpt, 85 problems / 17 categories.

SKILLS

Languages: Python, Java, JavaScript/TypeScript, SQL

Back End & Cloud: Node.js, Express.js, FastAPI, Flask, REST APIs, microservices, Pub/Sub, GCP Cloud Run/Functions/Document AI/IAM, Docker

Front End & Mobile: React, React Native (Expo), HTML/CSS, responsive design, Tailwind, Next.js

Data & AI: PostgreSQL, MySQL, MongoDB, SQLite, Prisma, pandas, NumPy, PyTorch, PyTorch Geometric, scikit-learn, LLM APIs, RAG, prompt engineering

Optimization: Mixed-integer linear programming, OR-Tools CP-SAT, genetic algorithms, simulated annealing, tabu search, Bayesian hyperparameter optimization